

A hierarchical model for analyzing multisite individual-level disease surveillance data from multiple systems

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Abstract

Passive surveillance systems are widely used to monitor diseases occurrence over wide spatial areas due to their cost-effectiveness and integration into broadly distributed healthcare systems. However, such systems are generally associated with imperfect ascertainment of disease cases and with heterogeneous capture probabilities arising from factors such as differential access to care. Augmenting passive surveillance systems with other surveillance efforts provides a way to estimate the true number of incident cases. We develop a hierarchical modeling framework for analyzing data from multiple surveillance systems that allows for individual-level covariate-dependent heterogeneous capture probabilities, and borrows information across surveillance sites to improve estimation of the true number of incident cases. Inference is carried out via a two-stage Bayesian procedure. Simulation studies illustrated superior performance of the proposed approach with respect to bias, root mean square error, and coverage compared to a model that does not borrow information across sites. We applied the proposed model to data from three surveillance systems reporting pulmonary tuberculosis (PTB) cases in a major center of ongoing transmission in China. The analysis yielded bias-corrected estimates of PTB cases from the passive system and led to the identification of risk factors associated with PTB rates, as well as factors influencing the operating characteristics of the implemented surveillance systems.

KEYWORDS

Bayesian hierarchical modeling, heterogeneous capture probabilities, individual-level disease surveillance

1 | INTRODUCTION

Disease surveillance systems that monitor disease distribution and detect outbreaks are critical for developing, implementing, and evaluating public health programs.

So-called “passive” surveillance systems obtain incident case information through reports filed by healthcare facilities, including hospitals, clinics, and laboratories (Hadorn & Stärk, 2008). Data from passive surveillance systems have become more readily available in many parts of the

world due to increased capabilities of electronic reporting (Simonsen *et al.*, 2016; Huff *et al.*, 2017; Wang *et al.*, 2019). Although passive surveillance systems are generally more cost-effective and can be established over large geographic areas, they are limited in that they only detect cases that present to the healthcare system (Hadorn & Stärk, 2008). The probability of detection may also vary with both individual-level factors (eg, age, sex, household income, health literacy, and severity of symptoms) and population- or system-level factors (eg, availability of healthcare resources and reporting infrastructure) (Gibbons *et al.*, 2014; Peixoto *et al.*, 2020).

This paper focuses on addressing case underascertainment of a passive disease surveillance system operating over large geographic areas with heterogeneous populations. This is accomplished by leveraging additional data from “active” surveillance systems designed to periodically screen for disease in target populations, often utilizing highly sensitive, standardized diagnostic methods (Jamison *et al.*, 2006). Specifically, suppose an active surveillance system operates in a subregion within the passive system’s catchment area, then a capture-recapture (CRC) data set can be formed by identifying cases that are captured by both or either one of the systems. Analyzing the degree of overlap in observed cases between systems allows one to estimate system-specific capture probabilities and infer total cases (Gibbons *et al.*, 2014).

In this paper, we describe an approach for analyzing CRC data obtained by linking passive and active surveillance systems across multiple sites. The overlap between passive and active systems is used to estimate capture probability for the passive system, which allows us to bias-correct the number of cases reported by the passive system. The work is motivated by the increasingly observed scenario whereby more than one surveillance system collects individual-level case data. Our motivating application in China, for instance, comprises multiple passive and active surveillance systems that collect reports on infectious disease incidence. Our unified modeling framework aims to: (1) examine factors associated with individual-level, system-specific capture probabilities; (2) examine factors associated with true disease incident cases accounting for surveillance bias across sites; and (3) estimate true incident cases by bias-correcting data from a passive surveillance system.

Several key assumptions are necessary in the analysis of CRC data (Otis *et al.*, 1978; Lyles *et al.*, 2021). Independence between surveillance systems at the population level is a common assumption imposed in early statistical methods, including the well-known Lincoln-Petersen estimator and the bias-corrected Chapman estimator (Petersen, 1896; Lincoln, 1930; Chao *et al.*, 2001). Numerous statistical methods have been proposed to relax

the independence assumption, which are easily violated in the context of human disease surveillance (Chao *et al.*, 2001). For example, the widely used log-linear model in epidemiological studies allows for dependency structures between surveillance systems by incorporating interaction terms (Cormack, 1989). In addition, a large class of mixture models has been proposed that permit capture probabilities to vary across surveillance systems and individuals. Examples include logistic-normal models, beta-binomial models, latent class models, and models based on conditional capture probabilities (Coull & Agresti, 1999; Dorazio & Royle, 2003; Tounkara & Rivest, 2015; Farcomeni, 2016).

Geo-referenced CRC data have motivated methods that account for data collected at multiple locations, mostly in ecological applications. The N -mixture model proposed by Royle (2004) uses a Poisson distribution to model unobserved true animal population sizes at different spatial locations, which are treated as nuisance parameters and marginalized out of the likelihood. Recently, Li *et al.* (2020) extended the N -mixture model under a Bayesian spatial hierarchical framework to estimate disease incident cases at locations in which one or multiple surveillance systems operate. One major limitation of existing N -mixture approaches is that, while they allow capture probabilities to vary across surveillance systems and locations, they cannot account for individual-level heterogeneity in capture probability.

Our proposed approach combines an individual-level binomial mixture (BM) model, which allows for covariate-dependent capture probability, with a spatial Poisson regression model for site-specific true incident cases. Unlike previous N -mixture models for multisite CRC data, our approach treats the true number of incident cases as parameters of interest as part of hierarchical model. Inference and estimation are carried out via a Bayesian two-stage procedure to address missing covariate information on individuals not captured by any system. To our best knowledge, this is the first statistical framework that can accommodate both covariates for modeling spatially-varying incident cases at the aggregate level and covariates for modeling heterogeneity in capture probability at the individual level.

The rest of the paper is organized as follows. In Section 2, we introduce the motivating pulmonary tuberculosis (PTB) surveillance data from a major center of ongoing transmission in China. In Section 3, we describe the proposed statistical framework and the corresponding two-stage Bayesian estimation procedure. In Section 4, we present simulation studies for evaluating the performance of the proposed framework. The simulation results highlight the benefits of supplementing the BM model with a spatial process model and adding an additional

active system for estimating true incident cases. In Section 5, the proposed approach is applied to the PTB surveillance data, and we report estimated total PTB incident cases, and factors associated with system-specific capture probabilities and PTB rates. Finally, we conclude with a discussion of future methodological directions in Section 6.

2 | MOTIVATING APPLICATION AND DATA

In China, the National Infectious Disease Reporting System (NIDRS) is a passive, country-wide, web-based infectious disease reporting network that collects case data on reportable diseases. Complementary to NIDRS, cross-sectional prevalence surveys are conducted periodically (eg, every 5 years) for select infections, and provinces implement annual sentinel surveillance surveys in select locations. Hereafter, we refer to these three surveillance systems as **S1**, **S2**, and **S3**, respectively.

We analyzed linked **S1**, **S2**, and **S3** PTB data in Sichuan, a province with a high burden of PTB and a population of more than 80 million people (Yang *et al.*, 2008). The passive NIDRS (**S1**) has been operating in all 181 counties of Sichuan since 2004, and more than 71,000 PTB cases were reported in 2010 in the province. We supplemented **S1** data with data from a cross-sectional PTB prevalence survey (**S2**) conducted in 2010 for surveying PTB among populations over 15 years of age at the community level, as well as from sentinel surveillance (**S3**) conducted in nine select communities each year from 2012 to 2016 targeting populations over 15 years of age. The total number of PTB cases reported by these three surveillance systems represents an underestimation of the true number in Sichuan because these three surveillance systems cannot enumerate all PTB cases. Potential factors associated with underestimation including that **S1** only captures PTB cases who had encounter at health facilities covered by **S1**, and **S2** and **S3** (active systems) only capture cases at subregions of Sichuan. One goal of the current analysis is to examine individual and area-level covariates that may be associated with system-specific capture probability.

We obtained deidentified data linked between **S1** and **S2**, and between **S1** and **S3**, using a record linkage approach described elsewhere (Li *et al.*, 2019). **S1** data from 2009 to 2010 were linked to **S2** data, and **S1** data from 2011 to 2016 were linked to **S3** data. The aggregated ascertainment histories of the **S1**–**S2** and **S1**–**S3** linked data are summarized in Table 1. Figure 1 shows the community sites where both **S1** and **S2**, or **S1** and **S3**, were linked. We note that **S2** and **S3** do not temporally over-

TABLE 1 PTB cases ascertainment by **S1** and **S2** in 2009–2010, and by **S1** and **S3** in 2011–2016, in the study region

Captured in S1	2009–2010		2011–2016	
	Captured in S2 Yes	No	Captured in S3 Yes	No
Yes	84	37	85	155
No	117	?	172	?

lap during the study period. Our study included a total of 58 reporting sites with linked data: 24 sites that yielded **S1**–**S2** linked data, and 34 sites that yielded **S1**–**S3** linked data.

Individual-level case characteristics (eg, age, sex) were recorded for each individual by all three systems. Population-level risk factors of PTB and geographical information available for each surveillance site included annual gross domestic product per capita (GDP) from 2009 to 2016; prefecture-level average area per capita in 2010, a measurement of crowding (average area); county-level proportions of population >65 years old in 2010 (percent of elderly); longitude; latitude; and elevation. Summary statistics of individual-level and county/prefecture-level variables are presented in Web Tables 1 and 2.

3 | METHOD

3.1 | Model specification

Consider a passive surveillance system that operates at S sites. Let N_s denote the unobserved true number of incident cases at site s , and the total number of incident cases in the study region is given by $N = \sum_{s=1}^S N_s$. Among these S sites, a supplementary active surveillance system only occurs at K sites with $K < S$. Without loss of generality, we assume the first K sites have both passive and active surveillance data. We refer to these first K sites as linked sites, and the other $S - K$ sites as unlinked sites. For $i = 1, \dots, N_s$, $s = 1, \dots, S$, let y_{1is} denote the capture indicator of i th individual for the passive system. Similarly, for $i = 1, \dots, N_s$, $s = 1, \dots, K$, let y_{2is} denote the capture indicator for the active system.

We assume $Y_{tis} \sim \text{Bin}(1, p_{tis})$ for $t = 1, 2$ with density function $p_{tis}^{y_{tis}}(1 - p_{tis})^{(1-y_{tis})}$, and p_{tis} is interpreted as the capture probability of system t for individual i at site s . At the K linked sites, individual-level capture history is modeled using a BM model (Dorazio & Royle, 2003; Royle, 2004; Tounkara & Rivest, 2015). Assume Y_{1is} and Y_{2is} are independent given p_{1is} and p_{2is} , and because individuals with capture history $y_{1is} = 0$ and $y_{2is} = 0$ are not observed,

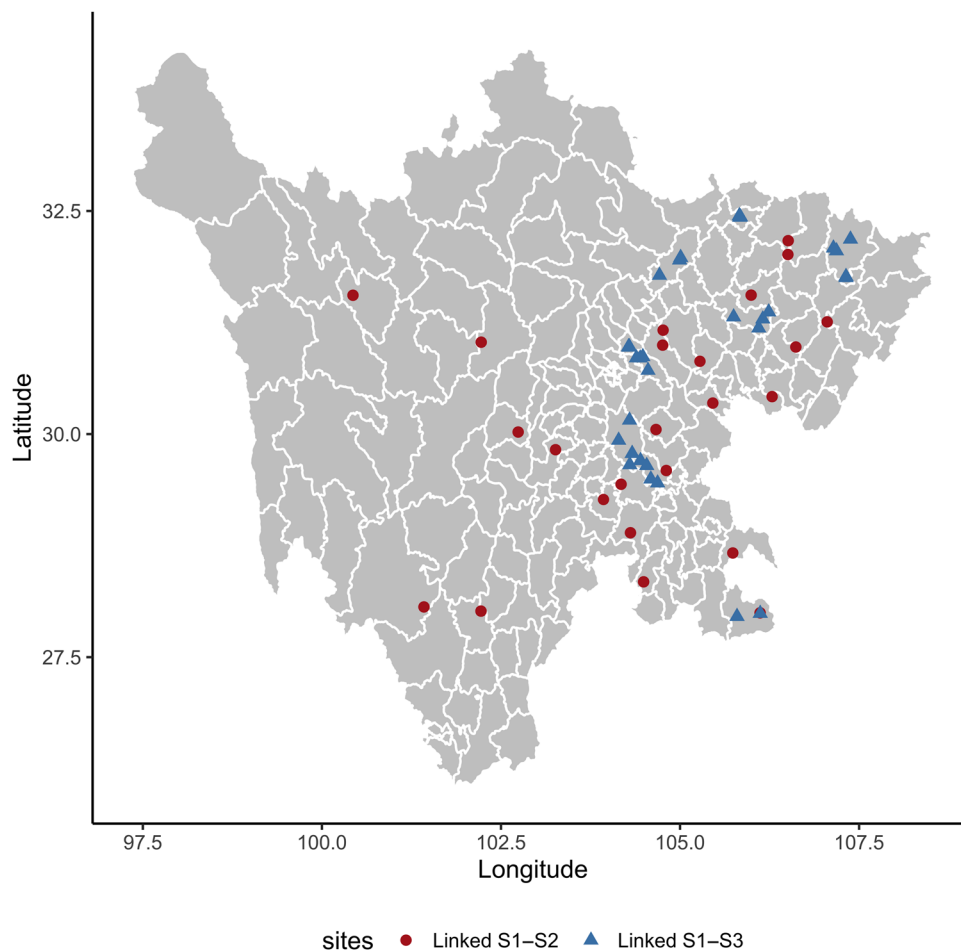


FIGURE 1 Counties within Sichuan Province where S1 and S2, or S1 and S3 surveillance are active. This figure appears in color in the electronic version of this article, and any mention of color refers to that version

the data likelihood of the capture history for individual i at the linked site s has a conditional joint density (Finney, 1947) given by

$$P(Y_{1is} = y_{1is}, Y_{2is} = y_{2is} | y_{1is} = 1 \text{ or } y_{2is} = 1) = \frac{p_{1is}^{y_{1is}} (1 - p_{1is})^{(1-y_{1is})} p_{2is}^{y_{2is}} (1 - p_{2is})^{(1-y_{2is})}}{1 - (1 - p_{1is})(1 - p_{2is})}.$$

To allow for individual heterogeneity in capture probabilities p_{1is} and p_{2is} , logit models are used to introduce individual-level covariates. In addition to linking active surveillance data and passive surveillance data, we further model the unobserved incident cases N_s as a Poisson process for all sites where any surveillance system operates, $s = 1, \dots, K, K + 1, \dots, S$. Our final proposed model is given by following hierarchical model that includes logistic regressions for individual-level capture probabilities and a spatial Poisson regression for site-specific incident cases.

For linked sites, $s = 1, \dots, K$:

$$\begin{aligned} P(Y_{1is} = y_{1is}, Y_{2is} = y_{2is} | y_{1is} = 1 \text{ or } y_{2is} = 1) &= \frac{p_{1is}^{y_{1is}} (1 - p_{1is})^{(1-y_{1is})} p_{2is}^{y_{2is}} (1 - p_{2is})^{(1-y_{2is})}}{1 - (1 - p_{1is})(1 - p_{2is})}, \\ \log \left\{ \frac{p_{1is}(\beta_1)}{1 - p_{1is}(\beta_1)} \right\} &= \mathbf{x}_{1is}^T \beta_1, \quad \log \left\{ \frac{p_{2is}(\beta_2)}{1 - p_{2is}(\beta_2)} \right\} \\ &= \mathbf{x}_{2is}^T \beta_2, \quad i = 1, \dots, n_s, \\ n_s &= \sum_i^{N_s} I(y_{1is} = 1 \text{ or } y_{2is} = 1). \end{aligned} \quad (1)$$

For unlinked sites, $s = K + 1, \dots, S$:

$$n_s = \sum_{i=1}^{N_s} I(y_{1is} = 1). \quad (2)$$

For all sites, $s = 1, \dots, S$:

$$\begin{aligned} N_s | \lambda_s(\alpha, \epsilon_s) &\sim \text{Poisson}(\lambda_s(\alpha, \epsilon_s)), \\ \log\{\lambda_s(\alpha, \epsilon_s)\} &= \mathbf{z}_s^T \alpha + \epsilon_s, \end{aligned} \quad (3)$$

where n_s is number of unique disease cases identified by surveillance systems at site s , $\mathbf{x}_{1is}$ and $\mathbf{x}_{2is}$ denote vectors of individual and location-specific covariates, such as demographic and residence information with corresponding system-specific coefficients β_1 and β_2 , $I(\cdot)$ denotes the indicator function, $\lambda_s(\alpha, \epsilon_s)$ is the expected number of incident cases at site s , $\mathbf{z}_s$ is a vector of risk factors for the disease of interest with regression coefficient α , and spatial random effects ϵ_s follow a mean-zero Gaussian process with an exponential covariance function. Specifically, $(\epsilon_1, \dots, \epsilon_S)^T \sim \text{MVN}(0, \Sigma)$, $\Sigma_{s,s'} = \sigma^2 \exp(-\frac{D_{s,s'}}{\phi})$, where $D_{s,s'}$ denotes the valid distance metric between site s and s' , and σ^2 and ϕ represent marginal variance and range parameter in exponential covariance function respectively.

3.2 | A two-stage Bayesian procedure for inference

Our goal is to obtain estimates of total cases N and site-specific cases N_s in the study region correcting for underascertainment of the surveillance system. The main difficulty in estimation is that site-specific cases N_s do not appear in the individual-level capture history likelihood. Hence, posterior samples of N_s cannot be directly obtained from specifying a full conditional distribution that includes contributions from both the BM model and the spatial Poisson model. We propose a Bayesian two-stage estimation procedure using Markov chain Monte Carlo (MCMC) algorithm for the hierarchical model. In the first stage, model coefficients β_1 and β_2 are estimated with the CRC data at the K linked sites. Given estimates of capture probabilities of observed individuals, posterior distributions of N_s are obtained using the Horvitz–Thompson (H–T) estimator. In the second stage, the estimated N_s and their associated uncertainties are treated as the observed data likelihood for the spatial Poisson regression. Furthermore, a simulation-based procedure based on posterior predictive distributions of the capture history is developed to account for using the H–T estimator in first-stage estimation. Details of the proposed two-stage estimation procedure are presented in next three sections.

3.3 | First-stage estimation

First-stage estimation is based on the BM model with CRC data formed by augmenting the passive system with the active system at K linked sites. The likelihood of β_1 and β_2 can be written

as

$$L(\beta_1, \beta_2; \{\mathbf{y}_{1,s}, \mathbf{y}_{2,s}\}_{s=1}^K) = \prod_{s=1}^K \prod_{i=1}^{n_s} \frac{p_{1is}^{y_{1is}} (1-p_{1is})^{(1-y_{1is})} p_{2is}^{y_{2is}} (1-p_{2is})^{(1-y_{2is})}}{1 - (1-p_{1is})(1-p_{2is})}, \quad (4)$$

where $\mathbf{y}_{t,s} = \{y_{tis}, i = 1, \dots, n_s\}$ for $t = 1, 2$ denotes individual capture history at site s . The posterior distributions of β_1 and β_2 are sampled by Metropolis–Hastings (M–H) algorithms. With posterior samples of model coefficients, the posterior distributions of individual-specific capture probabilities can be obtained by applying inverse logit transformation to linear combinations of posterior samples of model coefficients. Specifically, for the active system, the posterior distribution of p_{2is} is given by $\text{logit}^{-1}(\mathbf{x}_{2is}^T \beta_2)$, $i = 1, \dots, n_s, s = 1, \dots, K$. For the passive system, capture probabilities can be obtained for both linked and unlinked sites. In other words, p_{1is} is computed for each individual who has been identified by the passive system, $i = 1, \dots, n_s, s = 1, \dots, S$.

The H–T estimator of N_s is given by

$$N_s^* = \sum_{i=1}^{n_s} 1/q_{is}, \quad q_{is} := \begin{cases} 1 - (1-p_{1is})(1-p_{2is}) & \text{for } s = 1, \dots, K \\ p_{1is} & \text{for } s = K+1, \dots, S. \end{cases} \quad (5)$$

Simply applying posterior samples of p_{1is} and p_{2is} to Equation (5) will provide conditional posterior samples of N_s^* assuming known individual capture history, p_{1is} , and p_{2is} . We denote the desired first-stage marginal posterior distribution as $[N_s^* | \mathbf{h}_s]$, where $\mathbf{h}_s = \{\mathbf{y}_{1,s}, \mathbf{y}_{2,s}\}$ for linked sites and $\mathbf{h}_s = \mathbf{y}_{1,s}$ for unlinked sites. To appropriately propagate statistical uncertainties in estimating coefficients in the BM model to the H–T estimator (ie, generate samples from $[N_s^* | \mathbf{h}_s]$), we draw a random sample from the limiting distribution of the H–T estimator $N(\frac{1}{q_{is}}, \frac{1-q_{is}}{q_{is}^2})$ for each posterior sample of q_{is} (Huggins, 1989). This additional sampling step approximates the uncertainty in N_s^* with known capture probabilities. As a result of this simulation-based procedure, we obtain posterior samples which appropriately quantify uncertainties associated with using the H–T estimator. Finally, a point estimate of N_s denoted as $\hat{N}_s$ is obtained by computing the posterior mean; 95% credible intervals are obtained by taking the 2.5% and 97.5% percentile of the resulting posterior samples. The posterior samples of N are obtained by summing up posterior samples of N_s . In summary, first-stage

estimation yields preliminary estimates of total and site-specific incident cases solely based on CRC data at linked sites.

3.4 | Second-stage estimation

Let $\hat{\mathbf{N}} = (\hat{N}_1, \dots, \hat{N}_S)^T$ denote the vector of preliminary estimates of site-specific incident cases obtained from first-stage estimation and $\mathbf{N} = (N_1, \dots, N_S)^T$ denote the vector of true site-specific incident cases. To connect results from first-stage estimation to the Poisson log-linear model, we first note that the target joint posterior distribution of $\Theta = \{\mathbf{N}, \alpha, \sigma^2, \phi\}$ is

$$\begin{aligned} [\Theta|\hat{\mathbf{N}}] \propto & [\hat{\mathbf{N}}|\mathbf{N}, \{\mathbf{h}_s\}_{s=1}^S] \times [\mathbf{N}|\lambda] \times [\lambda|\alpha, \sigma^2, \phi] \times [\alpha] \\ & \times [\sigma^2] \times [\phi] d\mathbf{h}_1 \dots d\mathbf{h}_S, \end{aligned} \quad (6)$$

where $\lambda = (\lambda_1, \dots, \lambda_S)^T$. To obtain posterior samples of Θ with M-H algorithms based on Equation (6) using results obtained from first-stage estimation, we apply two approximations. First, to approximate $[\hat{\mathbf{N}}|\mathbf{N}, \{\mathbf{h}_s\}_{s=1}^S]$, we use a multivariate normal distribution with mean $\mathbf{N}$ and variance-covariance matrix computed as the empirical variance of $\hat{\mathbf{N}}$ from first-stage posterior samples. The second approximation replaces the outer integration of capture history via an additional sampling step. This approximation shares the spirit of multiple imputation approach advocated by Gelman *et al.* (1995) used in Bayesian analysis.

We begin by generating posterior samples of Θ using Equation (6) while ignoring the outer integration (ie, assuming capture history $\mathbf{h}_s$ is known from observed CRC data). We refer to these posterior samples as naive posterior samples, which will underestimate uncertainties associated with posterior distribution of $\mathbf{N}$ because uncertainties of capture histories have not been incorporated. To approximate the outer integration of Equation (6), $\hat{\mathbf{N}}_s$ and estimated coefficients of BM models are used to impute multiple data sets of individual-level capture history. In other words, we repeatedly draw observations $\{y_{1is}, y_{2is}\}$ from their posterior predictive distributions. Each imputed data set is analyzed separately using the two-stage estimation procedure to generate naive posterior samples of $\mathbf{N}$. Finally, naive posterior samples of all model parameters are pooled across imputed data sets to approximate samples generated from the desired distribution presented in Equation (6). Full details of generating imputed data set are given in [Web Appendix A](#). R code to carry out the two-stage estimation procedure with toy data is provided in the supporting information.

3.5 | Prior distributions

We assign weakly informative priors for model parameters. The prior distribution for each element in model parameter vectors β_1 , β_s , and α are normal distributions. For spatial random effect σ^2 and range parameter ϕ , the prior distributions are Inverse-Gamma and Gamma distributions, respectively. All distribution parameters are specified at values that represent weakly informative priors.

4 | SIMULATION STUDIES

Simulation studies were conducted to evaluate the performance of the model proposed in Section 3 for analyzing individual-level capture histories from multiple sites where active surveillance systems are operating in a subset of sites. We focused on bias, root mean square error (RMSE), and 95% credible interval coverage for the total incident cases N . A variety of statistical models have been developed for analyzing CRC data. However, none of them are directly comparable to our proposed model, which allows for individual-level covariates in capture probabilities and borrows information across surveillance sites within a unified framework. Hence, our simulation studies aim to illustrate potential advantages of the proposed model to the most competitive method for individual-level CRC data, that is, the BM model. Implementation of the BM model can be viewed as the first-stage estimation of our model. We note that the BM model does not borrow information across sites, nor does it utilize site-level covariates for case rate because the likelihood does not involve the unobserved true number of cases. We also examined the impacts of ignoring the spatial dependency by specifying exchangeable location-specific random effects (ie, nonspatial model) and including multiple active systems.

4.1 | Simulation design

We generated site-specific incident cases N_s from a spatial Poisson process with log mean α and an exponential covariance function for 50 sites. For each simulated individual, capture history was generated from a Binomial distribution with capture probabilities determined by logit models for the passive and active systems. The logit models contained individual-level continuous covariates $X_1 \sim \text{Uniform}(-1, 1)$ and $X_2 \sim \text{Uniform}(-2, 2)$, respectively, with both regression coefficients set to 1. Since the logit models include two different covariates, two systems work independently at the population level. Finally, the CRC data were obtained at 10 randomly sampled sites by

keeping individuals who were captured by at least one system. For the other 40 unlinked sites, only the capture history provided by the passive system was recorded. We examined the following simulation scenarios: (1) high versus low mean capture probabilities of the two systems by varying intercepts of the logit models (β_{10} and β_{20}), (2) strong versus weak spatial dependence in log case rate via parameter ϕ , and (3) high versus low case rate controlled by log mean α . Smaller values for β_{10} , β_{20} , α , and ϕ correspond to smaller mean capture probabilities for both systems, smaller overall case rate, and weaker spatial correlation between site-specific incident cases. A total of eight simulation scenarios were designed and 100 data sets were generated for each scenario. To mimic the real-world situation, N and N_s were fixed for each scenario, but individual-level capture history changed across replications. Details of data generation procedure for the simulation studies can be found in [Web Appendix B.1](#).

4.2 | Comparison with one-stage BM model

The estimated mean bias, RMSE, and coverage of 95% credible intervals of N for different simulation scenarios from the BM model, the proposed model, and the proposed model using exchangeable random effects in the second-stage estimation are summarized in Table 2. In all scenarios, our proposed model outperforms the BM model in terms of mean bias and RMSE when the spatial dependency is either incorporated or not. For the BM model, a positive bias is observed across all scenarios, especially when capture probabilities are small. This is likely because the H-T estimator may result in estimates which are biased upward when estimated capture probabilities are supplied, the positive bias may be more evident when number of captured cases is small (ie, capture probabilities are small) (McDonald & Amstrup, 2001; Tounkara & Rivest, 2015). The bias reduction associated with the proposed model is most remarkable when capture probabilities are small for both systems. For example, the mean bias decreased from 139 to 11 under Scenario 1 when average capture probabilities for both systems are around 27%. Although our proposed model has larger mean bias in Scenario 6, the difference is negligible compared to the true N . This could be due to the large number of observed cases to sufficiently estimate N_s based on the BM model. The relative RMSE using BM model as reference indicates that our proposed model with or without correct specification of the spatial dependency reduced RMSE under all scenarios, particularly when spatial correlation is strong. Under most of scenarios, the proposed model correctly incorporating spatial dependency has better

performance in terms of RMSE compared to the proposed model using exchangeable random effects. In terms of the coverage, the 95% credible intervals constructed by all models achieved or were close to the nominal level under most of scenarios. These results support the validity of the proposed two-stage estimation procedure in conducting inference of N . Without implementing the simulation-based procedure used for approximating the integral in Equation (6), the 95% CI resulted in undercoverage as suggested by the simulation results presented in Web Table 3.

4.3 | Positive dependence between systems

The simulations above assumed the two surveillance systems have independent capture probabilities at the population level. We also examined the performance of the proposed model when two surveillance systems are positively correlated at the population level. The positive dependency between two systems are introduced by including a common covariate to the two logit capture probability models; details about this simulation study can be found in [Web Appendix B.2](#). The mean bias and RMSE summarized from this simulation study suggest that the proposed model still outperforms the BM model under all simulation scenarios (Web Table 5).

4.4 | Benefits of multiple active systems

Our motivating application involves three surveillance systems, where a passive system is linked to two active systems, resulting in two independent CRC data sets. To illustrate the benefits of additional linkage between passive and active surveillance systems, we extended our simulation study to the scenario where two active systems that do not have overlap are available. The data generation procedure is similar to the procedure outlined for the previous simulation study with following differences: (1) two sets of linked sites without spatial overlap were simulated; and (2) a total of three logit models including an intercept and a common covariate were specified for each system, separately. The simulated data resulted in 100 cases on average at each site, and averaged capture probability of systems 1 (passive), 2 (active), and 3 (active) are 0.27, 0.35, and 0.33, respectively. We simulated a total of 100 sites, 10 sites were randomly selected as sites where S1 and S2 are linked, and 20 other sites were randomly selected as sites where S1 and S3 are linked. Details about modifying the proposed model to analyze such data and the full parameters setup can be found in [Web Appendix C and B.3](#).

TABLE 2 Mean bias, RMSE, coverage of 95% credible intervals of N estimated from BM model, the proposed model, and the proposed model using exchangeable random effects when two systems are independent at the population level over 100 simulated data

Scenario	True <i>N</i>	BM model			Proposed model				Proposed model (Exchangeable random effects)				
		Mean		Coverage	Mean		Coverage	RMSE	Relative ^a RMSE	Mean		Coverage	Relative ^a RMSE
		bias	RMSE		bias	RMSE				bias	RMSE		
Low prevalence ($\alpha = 3.3$), weak spatial dependence ($\phi = 10$)													
1 Low capture rate ($\beta_{10} = \beta_{20} = -1$)	1550	139	0.90	362	11	0.91	336	0.93	-17	0.88	333	0.92	
2 High capture rate ($\beta_{10} = \beta_{20} = -0.5$)	1896	70	0.98	241	-13	0.98	199	0.83	-17	0.99	207	0.86	
Low prevalence ($\alpha = 3.3$), strong spatial dependence ($\phi = 30$)													
3 Low capture rate ($\beta_{10} = \beta_{20} = -1$)	1643	153	0.95	354	-29	0.93	261	0.74	57	0.96	284	0.80	
4 High capture rate ($\beta_{10} = \beta_{20} = -0.5$)	1265	78	0.95	199	-27	0.98	149	0.75	2	0.97	157	0.79	
High prevalence ($\alpha = 4.3$), weak spatial dependence ($\phi = 10$)													
5 Low capture rate ($\beta_{10} = \beta_{20} = -1$)	3659	116	0.96	435	35	0.95	409	0.94	-11	0.97	379	0.87	
6 High capture rate ($\beta_{10} = \beta_{20} = -0.5$)	4180	5	0.96	301	-16	0.98	282	0.94	-21	0.97	280	0.93	
High prevalence ($\alpha = 4.3$), strong spatial dependence ($\phi = 30$)													
7 Low capture rate ($\beta_{10} = \beta_{20} = -1$)	4616	108	0.94	462	-24	0.95	426	0.92	41	0.96	446	0.97	
8 High capture rate ($\beta_{10} = \beta_{20} = -0.5$)	3612	82	0.96	288	-11	0.98	257	0.89	43	0.99	264	0.92	

^aRelative RMSE is computed using BM model as reference.

Table 3 presents the averaged estimates, averaged posterior standard deviation, and RMSE of regression coefficients of the BM model from using two data sets and one of them. With both active systems being linked to the passive system, coefficients associated with those three systems can be estimated simultaneously. As we expect, estimates of coefficients are empirically unbiased regardless of the degree of the system linkage. However, RMSE was reduced when two active systems were analyzed together compared to analyzing data sets separately (S1–S2 and S1–S3), particularly for coefficients related to capture probabilities of S1, the system is linked to both active systems. For example, comparing analysis using only S1–S2 linked data to using all data, the RMSE reduced from 0.32 to 0.19 for the intercept. This can be explained by the fact that both S2 and S3 contribute to the data likelihood used for estimating coefficients associated with S1. We also observed that RMSE

of coefficients associated with systems 2 and 3 decreased when two data sets were used simultaneously. This reduction is likely because the precision of estimated coefficients associated with S1 improved, resulting in more accurate estimation of capture probability associated with the two other active systems. Finally, as a result of improved estimation for coefficients of the BM model, the estimation of total number of cases also improved in the second-stage estimation when additional active systems are available (Web Table 6).

5 | APPLICATION

The proposed framework was applied to bias-correct reported PTB cases from the widely distributed passive system (S1) in Sichuan in 2010. We modified the model to

TABLE 3 Mean bias, averaged posterior standard deviation (SD), RMSE of regression coefficients in the BM model over 100 simulated data under situation where the passive system is linked to two active systems

		S1-S2 and S1-S3 linked ^a			S1-S2 linked ^b			S1-S3 linked ^c			
		True value	Mean bias	Posterior SD	RMSE	Mean bias	Posterior SD	RMSE	Mean bias	Posterior SD	RMSE
S1	intercept	-1.75	-0.05	0.19	0.19	-0.07	0.33	0.32	-0.07	0.23	0.22
	slope	1	0.04	0.18	0.17	0.04	0.31	0.28	0.06	0.22	0.20
S2	intercept	-1.75	-0.02	0.26	0.25	-0.04	0.34	0.32	-	-	-
	slope	1.5	0.04	0.27	0.26	0.04	0.33	0.30	-	-	-
S3	intercept	-1.85	0.05	0.21	0.23	-	-	-	0.04	0.24	0.24
	slope	1.5	0.05	0.21	0.24	-	-	-	0.06	0.23	0.25

Note: S1, S2, and S3 refer to surveillance systems 1, 2, and 3, respectively.

^aUsing both S1-S2 and S1-S3 linked data to estimate coefficients; all coefficients can be estimated.

^bUsing only S1-S2 data to estimate coefficients; only coefficients associated with capture probabilities of S1 and S2 can be estimated.

^cUsing only S1-S3 data to estimate coefficients; only coefficients associated with capture probabilities of S1 and S3 can be estimated.

allow for two independent CRC data sets, that is, the S1-S2 and S1-S3 linked data sets. In addition to estimating PTB cases, operating characteristics of the three surveillance systems and risk factors of PTB were also explored using the proposed model. The final logit models and spatial Poisson log-linear model including at-risk populations as an offset were selected by retaining covariates with 95% CI excluding 0. In this data analysis, age group with age <40 and female were used as reference group; all continuous risk factors of PTB and geographical information (longitude, latitude, and elevation) were centered at their means computed across all sites.

Goodness of fit of the model was assessed by comparing several quantities summarizing observed data to their corresponding posterior predictive distributions. We used the following quantities to summarize observed PTB data collected in Sichuan. For the S1-S2 linked data, we compute the number of PTB cases captured by S1 and S2 alone at linked sites, denoted as n_{S_1} , n_{S_2} , respectively. Similarly, we compute the number of PTB cases captured by S1 and S3 alone and denoted as $n_{S_1}^*$, n_{S_3} . From Table 1, we have $n_{S_1} = 121$, $n_{S_2} = 201$, $n_{S_1}^* = 240$, and $n_{S_3} = 257$. We also computed the number of PTB cases captured by S1 alone for each of 181 counties. The posterior predictive distributions of those quantities were obtained by simulation. Specifically, we drew observed outcomes (eg, capture indicators of S1 at S1-S2 linked sites) from its predictive distribution and then computed posterior median and 95% posterior intervals. Ten thousand samples were generated and details of simulating from the corresponding posterior predictive distributions are provided in Web Appendix D.

Figure 2 displays observed and posterior predictive distributions for summary statistics of observed linked and unlinked data to assess model fit. We observed that posterior predictive distributions are centered around observed

TABLE 4 Results from the first-stage estimation applied to analyzing individual capture probabilities of PTB data

Systems	Covariates	Estimated ORs (posterior SD)	95% Credible intervals
S1	Baseline	0.76 (0.16)	(0.50, 1.12)
	Longitude	1.18 (0.11)	(0.98, 1.41)
	Latitude	0.80 (0.06)	(0.68, 0.92)
	Year	0.84 (0.04)	(0.77, 0.92)
	Gender		
	male versus female	1.37 (0.25)	(0.95, 1.92)
S2	Baseline	0.82 (0.30)	(0.41, 1.55)
	Age (years)		
	[40, 50) versus < 40	3.14 (1.78)	(0.99, 7.65)
	[50, 60) versus < 40	5.65 (3.98)	(1.53, 15.99)
	[60, 70) versus < 40	10.03 (7.17)	(2.67, 28.17)
	≥ 70 versus < 40	10.89 (13.18)	(2.06, 36.65)
S3	Baseline	0.45 (0.13)	(0.26, 0.75)
	GDP	1.33 (0.18)	(1.02, 1.73)
	Elevation	10.41 (6.83)	(2.84, 27.92)
	Age (years)		
	[40, 50) versus < 40	1.77 (0.59)	(0.88, 3.17)
	[50, 60) versus < 40	2.35 (0.77)	(1.19, 4.24)
	[60, 70) versus < 40	2.75 (0.84)	(1.46, 4.75)
	≥ 70 versus < 40	5.89 (2.52)	(2.50, 12.04)

values, which suggest that the final model fitted PTB data well.

The estimated odds ratios for system-specific capture probability are presented in Table 4. Capture probability of S1 increased with latitude, decreased over time, and was higher for male cases. Unlike S1, the capture probabilities of the cross-sectional active system (S2) tended to have higher capture of cases in older age groups. Similar

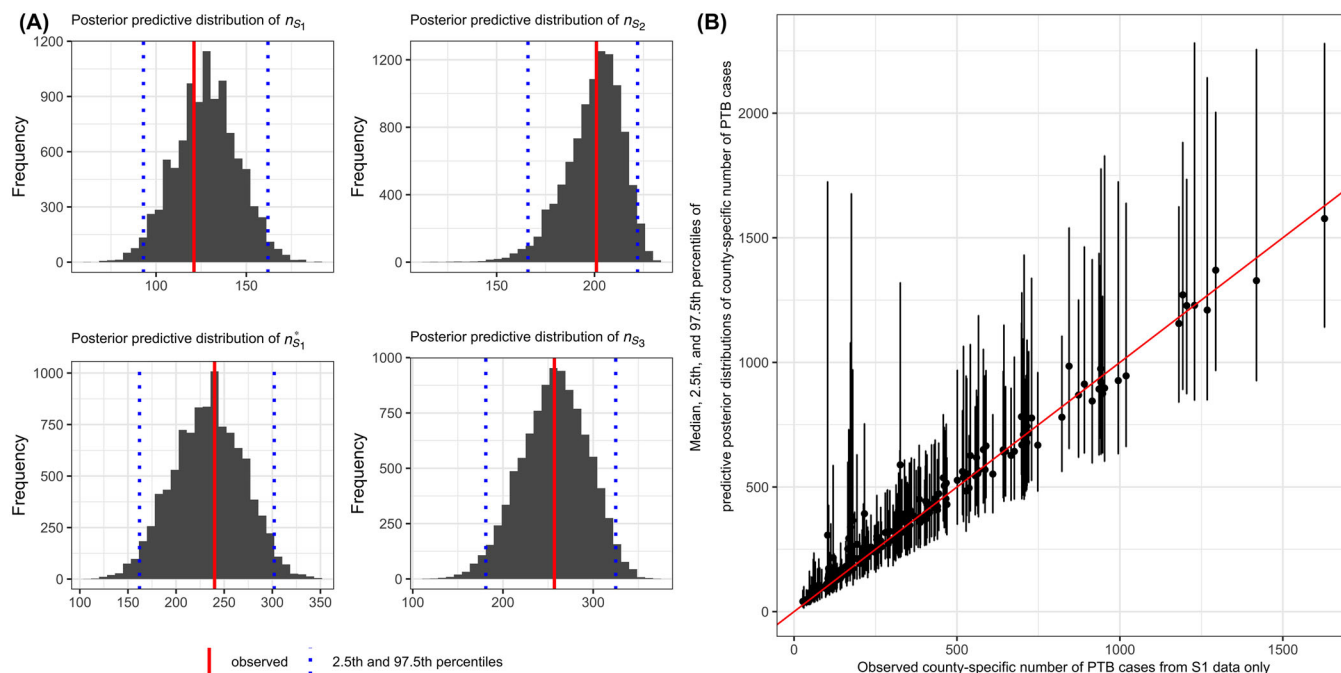


FIGURE 2 Observed and posterior predictive distributions for quantities summarizing observed data. This figure appears in color in the electronic version of this article, and any mention of color refers to that version

TABLE 5 Results from the second-stage estimation applied to analyzing PTB prevalence of PTB data

Covariates	Estimated RRs (posterior SD)	Posterior 95% CIs
Elevation	1.48 (0.17)	(1.17, 1.85)
Average area	0.94 (0.02)	(0.89, 0.99)
Percent of elderly	1.16 (0.05)	(1.06, 1.27)
σ^2	0.32 (0.08)	(0.21, 0.52)
ϕ	55.75 (15.49)	(32.88, 92.40)

age effects were also observed in the sentinel system (S3). In addition, S3 capture probability was strongly impacted by elevation. We also found that S3 had higher capture probability among patients who resided at locations with high GDP. The overall capture probabilities for each system were estimated to be 0.36 (95% CI: 0.32, 0.41) for S1, 0.73 (95% CI: 0.66, 0.79) for S2, and 0.37 (95% CI: 0.31, 0.43) for S3, respectively.

Table 5 presents the associations between several risk factors and PTB prevalence from the Poisson log-linear model. The effect of elevation on PTB prevalence (1.48 95%CI: 1.17, 1.85) indicates that areas with higher elevation tend to have higher PTB prevalence. Higher elevation areas in the study region are generally rural (most major urban areas reside in the low elevation plains) and exhibit lower socioeconomic status and poorer quality housing (Liu *et al.*, 2005; Wanyeki *et al.*, 2006). Proportion elderly and average area per capita (a measure of crowding)

were found to be significantly associated with PTB prevalence; and in fact they are known risk factors of PTB (Wanyeki *et al.*, 2006; Lönnroth *et al.*, 2009; Li *et al.*, 2019).

From our hierarchical model, the total number of PTB cases in Sichuan in 2010 is estimated to be 170,579 (95% CI: 138,180, 209,971), which is more than twice the number of cases observed in S1 alone. Figure 3 displays the maps of observed versus adjusted incident PTB cases from the passive system (S1) and the corresponding prevalence (per 1000 population). There is high spatial heterogeneity in PTB cases across the study region, and S1 suffers from substantial underascertainment. The prevalence estimates in Figure 3 help identify areas with high PTB prevalence that may be neglected if only unadjusted S1 data are considered. To better visualize differences spatially, Web Figures 1 and 2 present the estimated bias for the number of PTB cases and PTB prevalence associated with S1.

6 | DISCUSSION

In this paper, we propose a statistical framework that integrates a BM model with a spatial Poisson model to address underascertainment of a passive surveillance system when active surveillance systems data are also available to construct a CRC data set in subregions. To our best knowledge, this is the first time the commonly used BM model for individual-level CRC data and Poisson model for

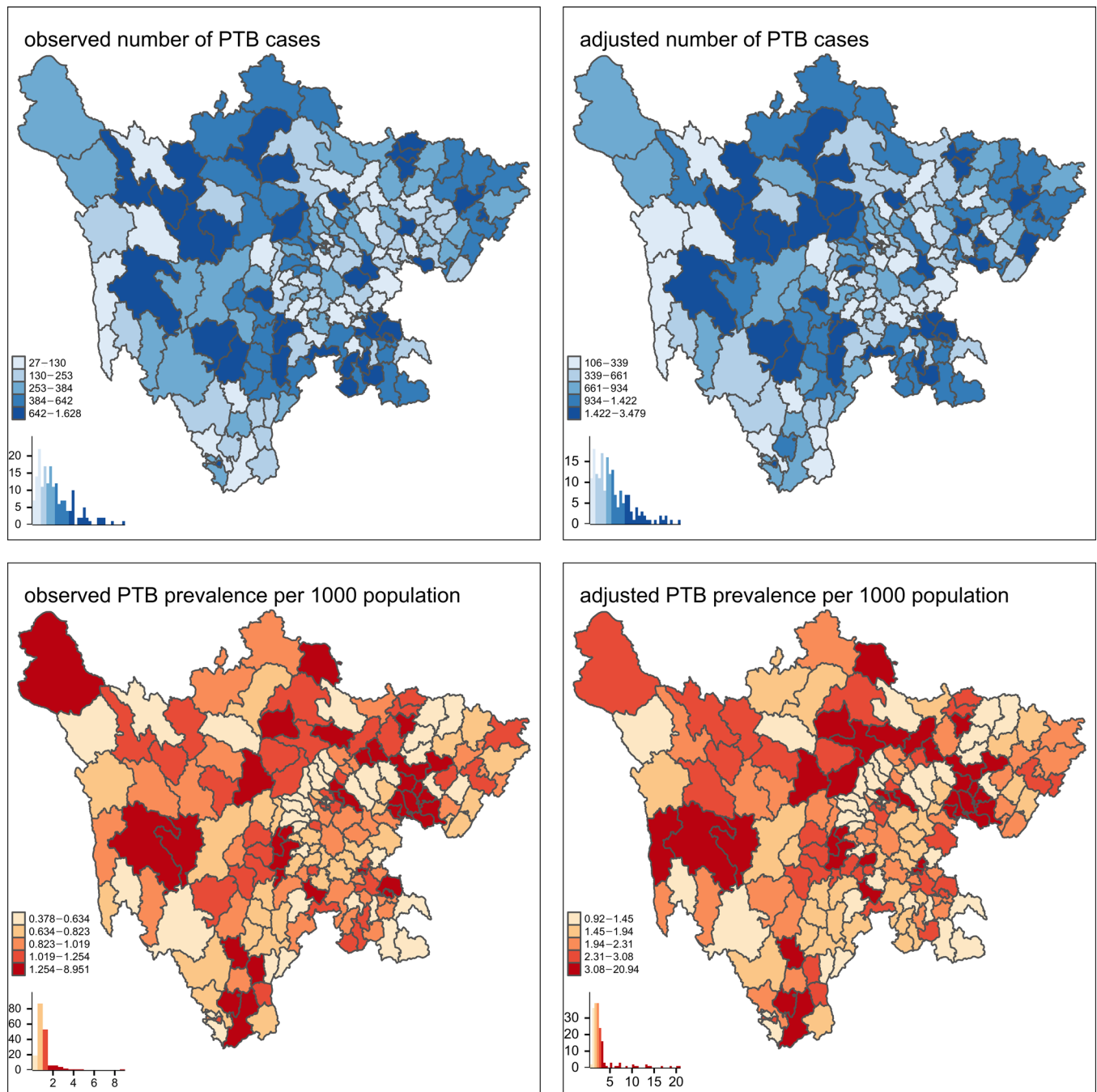


FIGURE 3 A map of observed/adjusted number of PTB cases, and observed/adjusted PTB prevalence per 1000 population in Sichuan in 2010 with county borders. This figure appears in color in the electronic version of this article, and any mention of color refers to that version

aggregate-level incident cases are employed jointly. Doing so allows us to incorporate individual-level covariates for capture probabilities, covariates for disease rates, and spatial dependence. A two-stage Bayesian estimation procedure is developed for conducting inference of the incident cases, while addressing missing covariate information for individuals not captured by any system. Compared to the methods which solely rely on CRC data, the additional latent process on N_s allows us to utilize data from locations without active surveillance.

Applying the proposed model to the study of PTB in the study region, we identified factors that affect the capture probabilities of three surveillance systems that monitors PTB, risk factors, and spatial pattern of PTB cases. Passive surveillance (S1) exhibited the lowest average capture probability and was impacted by geographic location, time period, and case sex. When appropriate, estimates of parameters that govern S1 capture probabilities can be used to bias-correct S1 PTB cases at locations and years without active surveillance. Surprisingly, on average, the

capture probability of sentinel surveillance (S3) was low (0.37) and was impacted by economic conditions, geographic location, and age. This low capture probability may indicate that this active surveillance effort was not well implemented. The factors identified as influencing capture probabilities may be considered as strategies are sought to strengthen public health surveillance in the study region.

Our proposed model can be viewed as a general hierarchical framework for correcting raw cases reported by passive surveillance systems when other samples of the epidemiological process of interest are available. Specifically, the first-stage model does not have to be restricted to the BM model presented in Section 3. Any model that provides site-specific estimation of incident cases based on CRC data, such as models that belong to the classes of mixture models, can be utilized. Our model can also be easily extended to dealing with the situation in which the targeted passive system can be augmented with multiple active surveillance systems, since the BM model used in the first-stage estimation is able to accommodate CRC data formed by multiple surveillance efforts (Toukara & Rivest, 2015). In light of this generality, one important future extension of the proposed model is to include consideration of unmeasured heterogeneity in capture probability. For example, to account for both measured and unmeasured heterogeneity, the logit-normal model proposed by Coull and Agresti (1999) is an appropriate model considered in the first-stage estimation.

The H-T estimator may result in unrealistically large estimation of site-specific incident cases when estimated individual-specific capture probabilities are close to zero. This is because individuals who have very low probabilities of being captured will contribute greatly to the estimation of N . These unrealistically large estimates may affect our second-stage estimation. This warrants a further direction to develop or incorporate an improved estimator that is robust to extremely small capture probabilities as a substitute for the H-T estimator. For example, one convention adopted in estimating average treatment effect with inverse probability weighting, is to restrict propensity scores within the range [0.1, 0.9] (Crump *et al.*, 2009).

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DATA AVAILABILITY STATEMENT

Data cannot be shared publicly because of restrictions that apply to protected health information (PHI). The data underlying the results presented in this paper are available from the Centers for Disease Control and Prevention, Chengdu, Sichuan, China, which can be reached at zxbsgs@scdc.cn or by phone 028-85580303. All other relevant data without personally identifiable information are within the paper.

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SUPPORTING INFORMATION

Web Appendices, Figures, and Tables referenced in Sections 2, 3.4, 4 and 5, along with code to illustrate the estimation procedure with toy data are available with this paper at the Biometrics website on Wiley Online Library.

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